The Galois module structure of Deligne-Lusztig varieties for GL_3

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Outline

Deligne–Lusztig varieties

Galois–module structures

Setup and Main results

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  - an isogeny F such that F^δ is a power of the Frobenius endomorphism for some $\delta \geq 1$
  - a Weyl group element $w \in W$.
- Furthermore, these varieties fit into a finite étale cover $\pi_w : \mathbf{Y}(w) \to \mathbf{X}(w)$ such that

$$\mathbf{X}(w) \simeq \mathbf{Y}(w)/\mathbf{T}^{wF}$$

i.e., $\mathbf{X}(w)$ is a quotient of $\mathbf{Y}(w)$ by the action of the finite group $\mathbf{T}^{wF}$ of rational points of a torus.

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- We also have that $\mathscr{X} := \mathbf{G}/\mathbf{B}$ is a projective variety.
- We cut out a locally closed subvariety $\mathbf{X}(w) \subseteq \mathbf{G}/\mathbf{B}$ of $\mathscr{X}$ called the **Deligne–Lusztig variety** defined by

$$\mathbf{X}(w) \simeq \{\, g\mathbf{B} \in \mathbf{G}/\mathbf{B} \mid g^{-1}F(g) \in \mathbf{B}\dot{w}\mathbf{B} \,\} \subseteq \mathscr{X}$$

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- Then $\mathbf{Y}(w) \to \mathbf{X}(w)$ is a $\mathbf{T}^{wF}$-torsor, where

$$\mathbf{T}^{wF} = \{\, t \in \mathbf{T} \mid \dot{w}F(t)\dot{w}^{-1} = t \,\}.$$

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Examples

· $\mathbf{G} = \mathrm{GL}_n \,/\, \mathbb{F}_p$; $\quad F = \mathrm{Frob}_p$; $\quad w = (123\cdots n) \in S_n \simeq W \quad \rightsquigarrow$ ***Split case***

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$$Y(w) \simeq \left\{(x_1, \ldots, x_n) \in \mathbb{A}^n_{\mathbb{F}_p} : (-1)^{n-1} \det \left(x_i^{p^{j-1}}\right)_{1\leq i,j\leq n}^{p-1} - 1 = 0\right\}$$

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- Specifically, for $n = 2$, we get one of the irreducible components of $Y(w)$ to be the famous **Drinfeld curve** $xy^p - x^p y = 1$.

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Theorem (Bonnafé–Rouquier, 2009)

The normal variety $\overline{\mathbf{Y}}(w)$ is equipped with an action of $\mathbf{T}^{\mathbf{w}F}$ extending the action on $\mathbf{Y}(w)$ and such that $\overline{\pi}_{\mathbf{w}}$ induces an isomorphism $\overline{\mathbf{Y}}(w)/\mathbf{T}^{\mathbf{w}F} \xrightarrow{\sim} \overline{\mathbf{X}}(w)$.

$$\begin{array}{ccc} \mathbf{Y}(w) & \hookrightarrow & \overline{\mathbf{Y}}(w) \\ \downarrow{\scriptstyle \pi_w} & & \downarrow{\scriptstyle \overline{\pi}_w} \\ \mathbf{X}(w) & \hookrightarrow & \overline{\mathbf{X}}(w) \end{array}$$

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- Große–Klönne studied logarithmic crystalline cohomology on smooth compactifications $\overline{\mathbf{X}}(w)$, with coefficients computing the rigid cohomology of the corresponding Deligne–Lusztig coverings $Y(w)$ [Gro07]. Wang studied the structure-sheaf cohomology of $\overline{\mathbf{X}}(w)$ [Wan25].

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GALOIS–MODULE THEORY

One of the central problems in the subject of "geometric" Galois module theory is as follows:

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- Let $\mathcal{F}$ be a $G-$equivariant coherent sheaf of $\mathcal{O}_Y-$modules
  Then, the sheaf cohomology groups $H^i(Y, \mathcal{F})$ are G-modules, and one asks if its equivariant Euler characteristic

$$\chi(Y, \mathcal{F}) := \sum_i (-1)^i [H^i(Y, \mathcal{F})] \in G_0(\bar{k}[G])$$

  can be calculated using a bounded complex of finitely generated free $k[G]-$modules.

- When $p = \operatorname{char} k \nmid |G|$, Maschke's theorem identifies $G_0(\bar{k}[G])$ with $K_0(\bar{k}[G])$, in which case we call this alternating class $\chi_G(Y, \mathcal{F})$ *the virtual Galois module*.

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- One would like to compute the class $\chi_G(Y, \mathcal{F})$ in $K_0(\bar{k}[G])$, or equivalently to determine all of its Brauer-character values.

- A basic question is whether this class is *virtually free*, that is, whether

$$\chi_G(Y, \mathcal{F}) \in \mathbb{Z} \cdot [\bar{k}[G]] \subseteq K_0(\bar{k}[G]).$$

- For a p-regular element $\tau \in G$, we define its **Lefschetz number** $L_{\mathrm{Br}}(\tau, Y, \mathcal{F})$ to be

$$L_{\mathrm{Br}}(\tau, Y, \mathcal{F}) := \sum_i (-1)^i \operatorname{BrTr}\big(\tau \mid H^i(Y, \mathcal{F})\big),$$

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- Since the Brauer character of the regular representation $\bar{k}[G]$ vanishes at every non-identity element of G, if $\chi_G(Y, \mathcal{F})$ **is virtually free**, then $L_{\mathrm{Br}}(\tau, Y, \mathcal{F}) = 0$ **for every** non-identity element $\tau \in G$.

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- We are interested in the Lefschetz numbers for the tamely ramified cover of compactified Deligne-Lusztig varieties $\overline{\mathbf{Y}}_\ell(w) \to \overline{\mathbf{X}}(w)$ for $\mathbf{G} = \mathrm{GL}_n$.
  i.e.,
  $$L(\tau, \overline{\mathbf{Y}}(w), \mathcal{O}_{\overline{\mathbf{Y}}_\ell(w)}) := \sum_i (-1)^i \mathrm{Tr}\Big(\tau | H^i\big(\, \overline{Y}_\ell(w), \mathcal{O}_{\overline{\mathbf{Y}}_\ell(w)}\,\big)\Big)$$
  where $\tau \in \mathrm{Gal}(\overline{\mathbf{Y}}(w) \,/\, \overline{\mathbf{X}}(w)) = T^{wF}$ and $\overline{\mathbf{Y}}_\ell(w)$ is an irreducible component of $\overline{\mathbf{Y}}(w)$.

Lefschetz–Riemann–Roch for surfaces

Let X be a smooth projective connected surface over $\bar{k}$, and let τ be a nonidentity automorphism of finite order prime to $p = \mathrm{char}(\bar{k})$. Its fixed locus is a disjoint union

$$X^\tau = \{P_1, \ldots, P_r\} \sqcup D_1 \sqcup \cdots \sqcup D_s$$

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The Lefschetz number is given by

$$L_{\mathrm{Br}}(\tau, X) := \sum_i (-1)^i \,\mathrm{BrTr}\big(\tau \mid H^i(X, \mathcal{O}_X)\big) = \sum_P a(P) + \sum_D b(D),$$

where

$$a(P) = \frac{1}{(1-\lambda_1)(1-\lambda_2)}, \qquad b(D) = \frac{1-g(D)}{1-\mu} - \frac{\mu D^2}{(1-\mu)^2}.$$

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where, λ_1 and λ_2 are the **eigenvalues** of the action of τ on the **tangent space** $T_P X$ at the point P, g_D is the genus of D, and D^2 is **the self-intersection number** of D, and μ is the eigenvalue of τ on **the normal bundle** of D.

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- the isogeny F:
  - $F = \mathrm{Frob}_p \quad \rightsquigarrow \quad$ *Split case*
  - $F(g) = J(g^{(p)})^{-t} J^{-1}$, where $J = \mathrm{antidiag}(1, 1, 1)$ and $-t$ denotes the inverse transpose $\quad \rightsquigarrow \quad$ *Non-split case*

Main Results (Split case)

Theorem (S.)

Choose $\mu \in \mathbb{F}_{p^3}^{\times}$ of order $p^3 - 1$, and set

$$\tau := \mathrm{diag}(\mu, \mu^p, \mu^{p^2}), \qquad N := p^2 + p + 1, \qquad \sigma := \tau^{p-1}, \qquad \bar{\zeta} := \mu^{p-1}.$$

Then τ generates $\mathbf{T}^{wF}$, while σ generates the stabilizer of each of the $(p-1)$-many irreducible components $\overline{Y}_\ell$ of the compactified cover $\overline{Y}$.

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Put $\widehat{\zeta} = [\bar{\zeta}] \in W(\mathbb{F}_{p^3})^{\times}$. Write $\overline{Y} = \bigsqcup_{\ell \in \mathbb{F}_p^{\times}} \overline{Y}_\ell$. For every such component,

$$L_{\mathrm{Br}}(\sigma, \overline{Y}_\ell(w)) = N(p+1)\,\frac{\widehat{\zeta}\,(\widehat{\zeta}^{p-1} - 1)}{(1-\widehat{\zeta})^2(1-\widehat{\zeta}^p)} + N\,\frac{1 + (p-1)\widehat{\zeta}}{(1-\widehat{\zeta})^2} + N\,\frac{\widehat{\zeta}^{2p}}{(1-\widehat{\zeta}^p)^2} \quad (1)$$

Furthermore, both the Lefschetz number in (1) and the value for the entire normal compactification are nonzero.

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Then, $\mathbf{T}^{wF} = \left\{ \mathrm{diag}(\zeta, \alpha, \alpha^{-p}) : \zeta^{p+1} = 1, \quad \alpha^{p^2-1} = 1 \right\} \simeq \mu_{p+1} \times \mu_{p^2-1}$.

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The normal compactification $\overline{Y}$ has exactly $p+1$ irreducible components $\overline{Y}_\ell$, and τ generates the stabilizer H of each of those components. Fix one of them and denote it by $\overline{Y}_\ell$. Set

$$c_0 = \frac{(p+1)(p-2)}{2}, \qquad c_u = \frac{(p+1)(3p-2)}{2}, \qquad c_v = \frac{(p+1)(3p-4)}{2},$$

which are all integers for every prime p. For this chosen component,

$$L_{\mathrm{Br}}(\tau, \overline{Y}_{\delta_0}(w)) = \frac{c_u\widehat{\xi} - c_0}{(1-\widehat{\xi})^2} - \frac{(p^3+1)\,\widehat{\xi}}{(1-\widehat{\xi})(1-\widehat{\xi}^p)} + \frac{\widehat{\xi}^p\,(c_v - c_0\widehat{\xi}^p)}{(1-\widehat{\xi}^p)^2} \tag{2}$$

Furthermore, both the Lefschetz number in (2) and the Lefschetz number for the entire normal compactification are nonzero.

Some explicit geometry

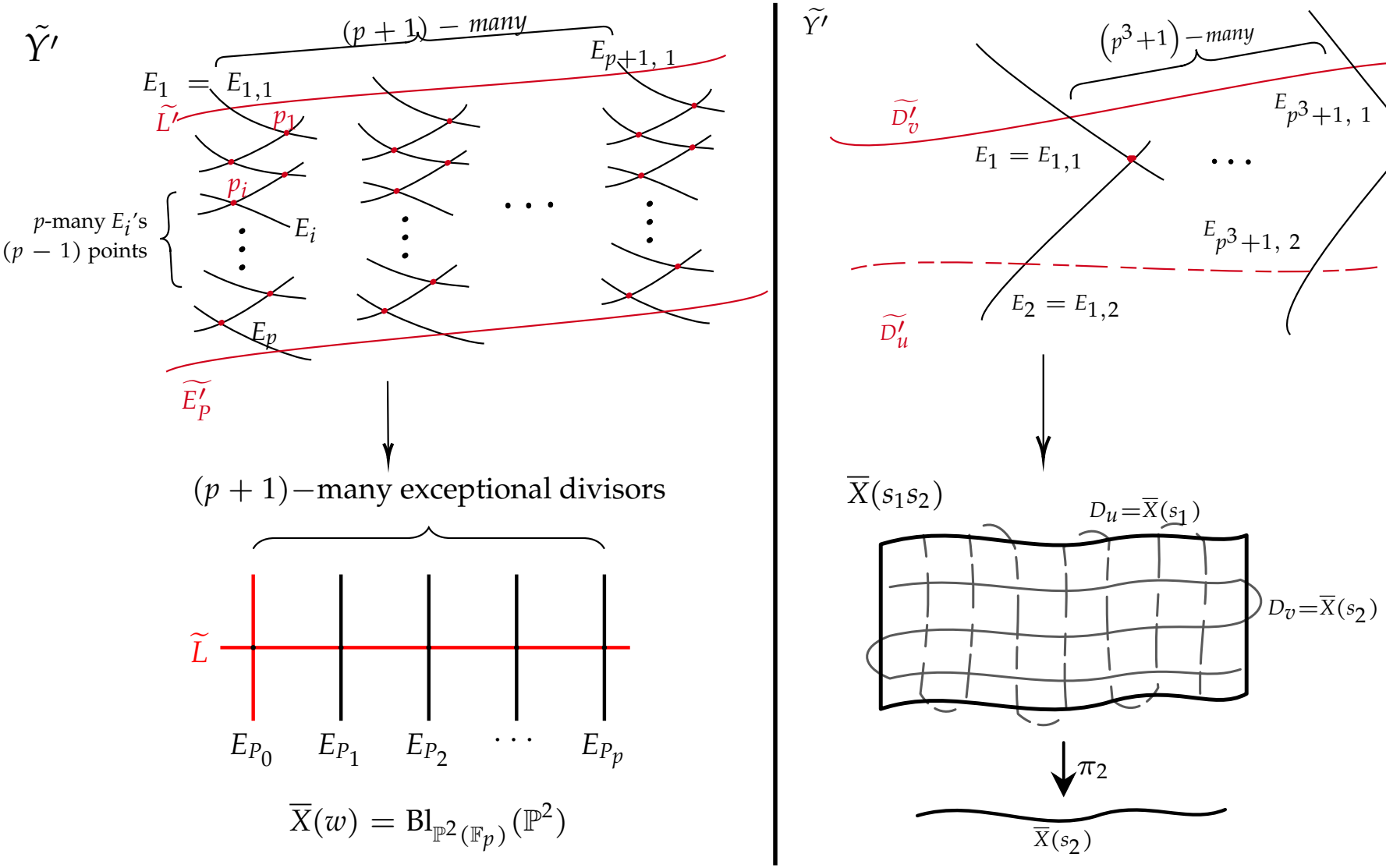


Figure: Resolutions of the split ($G^F = \mathrm{GL}_3(\mathbb{F}_p)$) case on the left and the non-split ($G^F = \mathrm{U}_3(\mathbb{F}_p)$) case on the right.

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